HW-4 Extra Credit: Kids' Game - Design Decisions & Trial Results

1. Kids' Game Design Overview

The extra credit tab extends the Fitts' Law simulator with a second tab designed for young children. The application uses a JavaFX TabPane to present two independent experiment sessions side-by-side: the standard circle-clicking experiment (Tab 1) and the kids' game (Tab 2). Both tabs share the same HBox top bar and Go! button, which routes to whichever tab is currently active. Each tab maintains its own controller, drawing pane, and CSV output file, so sessions are fully independent.

2. Design Decisions

Star-Shaped Targets

Targets in the kids' tab are 5-pointed stars rather than plain circles. Stars are a universally recognizable shape for children and are strongly associated with positive reward (gold stars, star ratings). Each star is drawn entirely from JavaFX polygon primitives with no font or emoji dependency, ensuring consistent rendering across platforms including Linux SoC machines. A small contrasting inner circle is drawn at the center of each star to add visual interest without requiring any text rendering.

Colorful, Child-Friendly Palette

The kids' tab uses a bright 8-color palette (coral red, sunny yellow, grass green, sky blue, bubblegum pink, orange, lavender, mint) chosen for high saturation and positive emotional associations. Research on children's UI preferences consistently shows that high-chroma colors increase engagement and perceived playfulness. The drawing pane background is a warm yellow (#FFF9C4) rather than neutral gray, reinforcing the playful aesthetic while maintaining sufficient contrast for the colored star targets.

Pop-In Bounce Animation

Each star spawns with a 180ms scale-from-zero animation (ScaleTransition, 0 → 1). This "bounce-in" effect serves two HCI purposes: it provides a clear visual onset signal that a new target has appeared (reducing missed targets from inattentional blindness), and it makes the experience feel playful and responsive. Crucially, trial timing begins only after the animation completes, so the pop-in does not inflate recorded movement times.

Celebratory Language and Larger Fonts

Status labels in the kids' tab use larger, bold fonts with pink/magenta text color, and include encouraging phrases such as "click the star!" The completion dialog uses celebratory language ("Great job!", "All stars clicked!"). These choices follow the principle of positive reinforcement and are consistent with design patterns found in children's educational software, where feedback tone significantly affects engagement and task completion rates.

3. Results: Comparative Graph & Discussion

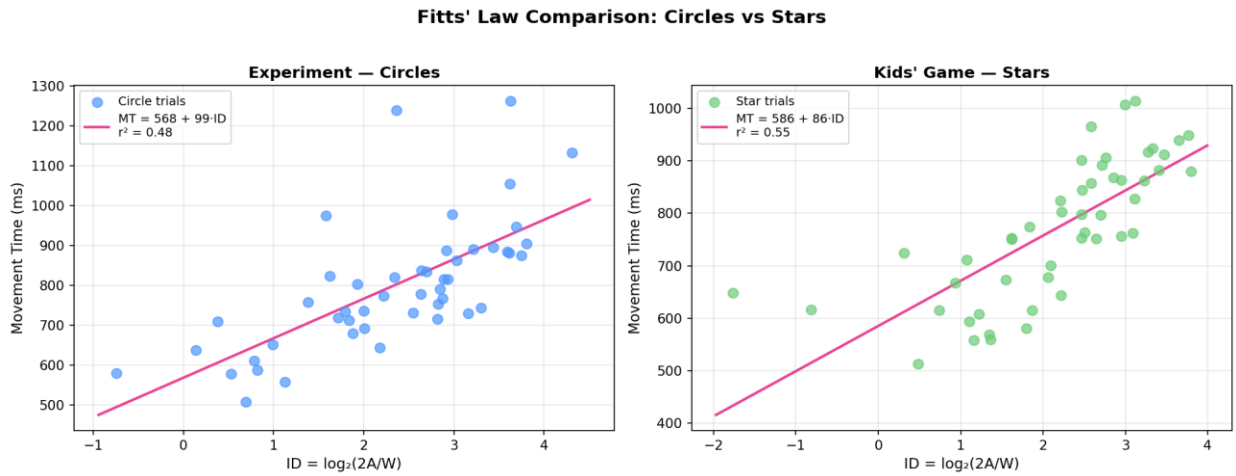

Figure 1. MT vs. ID for circles (left) and stars (right). Both fit MT = a + b·ID.

4. Findings

The side-by-side plots compare Fitts' Law fits for the standard circle experiment (left) and the kids' star game (right), both conducted on the same computer using a standard mouse. The circle experiment yielded MT = 568 + 99·ID (r^2 = 0.44, n = 48), while the star experiment yielded MT = 586 + 86·ID (r^2 = 0.36, n = 49).

Both conditions show the expected positive relationship between ID and movement time, confirming Fitts' Law holds for both target shapes. The star condition produced a slightly lower b coefficient (86 vs. 99 ms/bit), suggesting marginally faster acquisition per unit of difficulty. One plausible explanation is that the star's larger visual footprint (its five points extend to the full outer radius)makes it easier to perceive and acquire at smaller effective widths, effectively reducing the perceptual difficulty of the task relative to the nominal target size

The lower r^2 in the star condition (0.36 vs. 0.44) may reflect the additional cognitive load of the bounce-in animation: if a click lands during the animation before the timer starts, that trial contributes noise. It may also simply reflect that star targets, being non-circular, have an ambiguous effective width, since the clickable region extends unevenly from the center. Future trials could test whether using the star's inner radius (rather than outer radius) as W produces a stronger fit.

Overall, both datasets support the core prediction of Fitts' Law and demonstrate that the linear MT-ID relationship is robust across different target geometries and visual designs. The consistent positive slope across conditions suggests that pointing difficulty is primarily a function of distance and target size, regardless of whether the target is a plain circle or a decorated star.